Data Augmentation for SAR Sea Ice and Water Classification Based on Per-Class Backscatter Variation With Incidence Angle

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***Abstract*— Monitoring sea ice in polar regions is critical for understanding global climate change and supporting marine navigation. Recently, researchers started to utilize machine/deep learning methodologies to automate the separation of sea ice and open water in synthetic aperture radar imagery. However, this requires a large amount of reliably labeled training data. We here propose an augmentation routine for Sentinel-1 data, which incorporates physical principles of radar backscatter into the augmentation procedure. First, we apply an incidence angle (IA)-aware algorithm to segment Sentinel-1 images into separate clusters. We compute the corresponding slopes of backscatter intensity with the IA for each cluster. Second, the slopes are used as prior information to project labeled pixels and segments to different IAs and thus further enrich the labeled data. We then apply a simplified U-Net for pixelwise classification of Sentinel-1 images into sea ice or open water. The performance of our model is evaluated by visual inspection as well as comparison with an available product from the Chinese Academy of Sciences (CAS). The results indicate that the physics-based data augmentation improves the model performance compared to training with data without augmentation. The inferred ice edge is in line with the inference for other available datasets (CAS), but with a finer spatial resolution. Finally, we also found that the inference of our model is highly correlated with the visual interpretation of overlapping optical observations. Overall, the proposed methodology provides an alternative for the automated separation of sea ice/open water at fine spatial resolution.**

***Index Terms*— Data augmentation, deep learning, polar region, sea ice classification.**

I. Introduction

SEA ice is an important component of the Earth's climate system and a sensitive indicator of climate change. It is a threat to industrial offshore operations, as well as marine navigation in polar areas. Since 1979, Arctic sea ice coverage has decreased for all months of the year, with the largest decline in the summer and early fall. In September, at the end of melt season, the ice extent is now about half of the 1979 extent [1]. This significant sea ice loss has several climatological consequences. For instance, it leads to wetter European summers [2] and more extreme Northern Hemisphere winters [3]. Furthermore, it affects the Arctic marine ecosystems, which is, for example, indicated by the physiological adaptations of polar bears and narwhals [4], and it makes the Arctic more easily accessible for shipping and offshore operations [5]. Accurate and timely information about sea ice conditions is therefore crucial for safe Arctic operations, for improving weather and climate models, and it can be used to support policymakers in making local, regional, as well as global regulations to prevent deterioration of the Arctic ecosystem.

Sea ice extent (SIE) and sea ice concentration (SIC) are two main parameters to describe the state of the Arctic sea ice cover. At present, passive microwave radiometer (PMR) observations are the main data source for pan-Arctic mapping of SIE and SIC. For instance, researchers from the University of Bremen have developed an SIC product based on Advanced Microwave Scanning Radiometer 2 (AMSR-2) data, using an algorithm developed during the Arctic radiation and turbulence interaction study (ARTIST) [6]. Similarly, the bootstrap (BT) algorithm [7] is being employed by NASA to provide a near-real-time NOAA/NSIDC Climate Data Record of Passive Microwave Sea Ice Concentration (https://nsidc.org/data/g10016), utilizing observations since 1978.

Both of these datasets are widely used by researchers for sea ice climatology studies. However, the spatial resolution of the PMR sensors is coarse, on the order of several kilometers. This is not suitable for operational ice charting to support marine navigation, which requires continuous information on a much finer spatial resolution. Such information can only be reliably provided by spaceborne synthetic aperture radar (SAR) observations, which are independent of Sun illumination and cloud coverage. The weather-independent and reliable imaging capability makes SAR the main data source in national sea ice services around the world. At present, operational sea ice charts are produced manually by visual interpretation of the imagery by experienced sea ice analysts [8], a process that is both subjective and time-consuming. Furthermore, the manually drawn ice charts do not always

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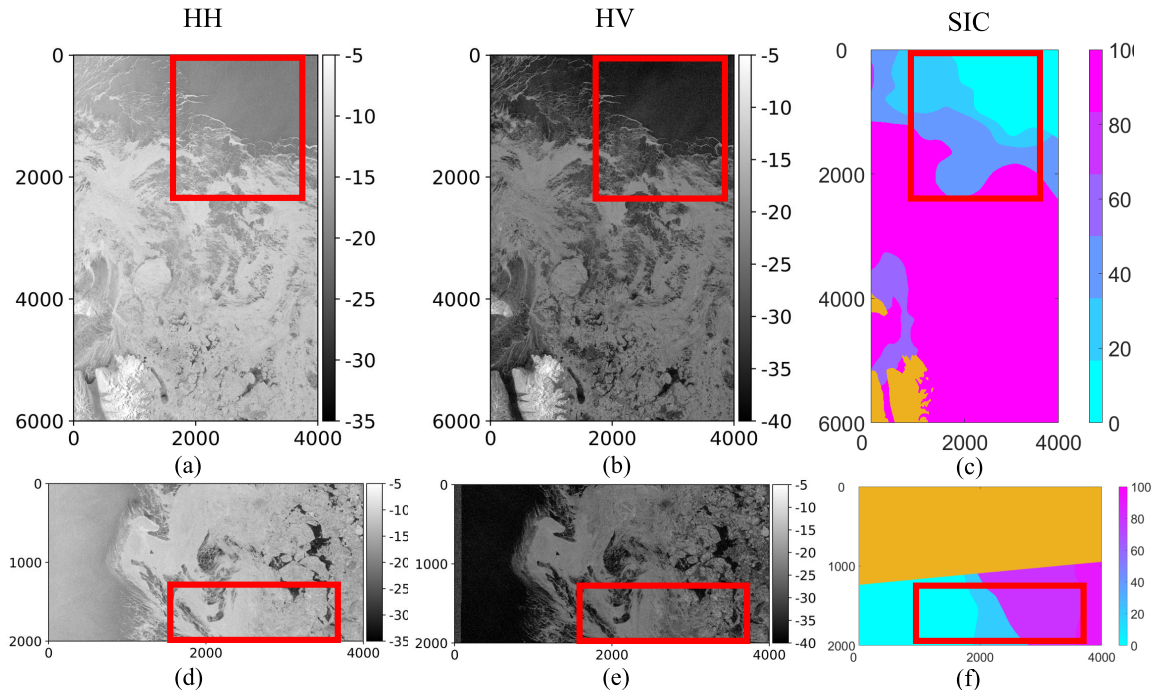

Fig. 1. (a) and (d) HH and (b) and (e) HV intensity (in dB) for selected regions of two S1 example images (scene NO.9 and 10 in Table I), together with (c) and (f) SIC from the corresponding DMI ice chart. Note the differences in spatial detail (indicated by red rectangles) between the original SAR images and the manually drawn ice charts. The SAR images contain more spatial information than what is captured in the ice chart. If the classifier is expected to reproduce these details, they must be included in the training.

reflect the fine pixel-level spatial detail of the underlying SAR imagery, as is, for example, shown in the ice charts produced by the Danish Meteorological Institute (DMI) in Fig. 1 (https://data.dtu.dk/articles/dataset/AI4Arctic_ASIP_Sea_Ice_Dataset_-_version_2/13011134).

Hence, a vast number of scholars have made efforts toward automating the ice charting process from SAR at the pixel level. For instance, Lohse et al. [9] utilized a Bayesian classifier for sea ice-type mapping based on Sentinel-1 HH and HV intensities, incidence angle (IA), and selected gray-level co-occurrence matrix (GLCM) texture features. Cristea et al. [10] proposed an unsupervised sea ice segmentation algorithm by integrating the target-specific IA-dependent intensity decay rates into a nonstationary statistical model.

With the recent advancement of artificial intelligence technology, researchers also started to utilize machine learning methods for sea ice mapping. Zakhvatkina et al. [11] utilized a support vector machine (SVM) classifier, based on Radarsat-2 HH and HV backscatter intensities as well as texture features, to separate sea ice and open water. Park et al. [12] developed a semiautomated sea ice classifier based on a random forest algorithm, using sea ice charts to extract training data. It is known that the input features used to train classification algorithms (e.g., backscatter intensities or texture features with selected parameter settings) affect the generalization and performance. Traditionally, these features are manually selected. In deep neural networks (DNNs), which are now being extensively explored also for sea ice applications, the networks automatically learn the optimal features through a hierarchy of computational layers. For instance, Khaleghian et al. [13] developed a modified convolutional neural network (CNN) architecture based on Visual geometric Group 16-layer (VGG-16) for binary sea ice versus open water classification. They also studied multiclass sea ice-type classification using the same architecture. Malmgren-Hansen et al. [14] proposed a CNN architecture for fusing Sentinel-1 SAR imagery and AMSR-2 PMR data for ice charting, while Stokholm et al. [15] used a modified U-Net architecture, where they included the SAR noise correction scheme developed by the Nansen Environmental and Remote Sensing Center (NERSC) [16]. Wang and Li [17] utilized multiple U-Net classifiers to map the Arctic sea ice cover from Sentinel-1 imagery and found a 5.55% difference compared with AMSR-2 SIC data.

Successful sea ice classification based on DNN algorithms requires a large number of high-quality training data with known class labels. Several such training datasets are publicly available. However, most of these datasets are based either on manual/visual analysis of the SAR imagery or on sea ice charts produced by the national ice services (which are eventually also based on manual analysis). Consequently, the class labels often lack detailed spatial information at the SAR pixel level. This is the case for the previously mentioned DMI dataset (Fig. 1) as well as, for example, for the UiT dataset (https://zenodo.org/record/4683174#.Yk7kVshBwuV) shown in Fig. 2. To increase the objectivity and robustness of DNN sea ice mapping algorithms, more high-quality training data at pixel-level resolution is required. While the manual selection of such training data is possible, it will be even more time-consuming than the generation of "traditional" training

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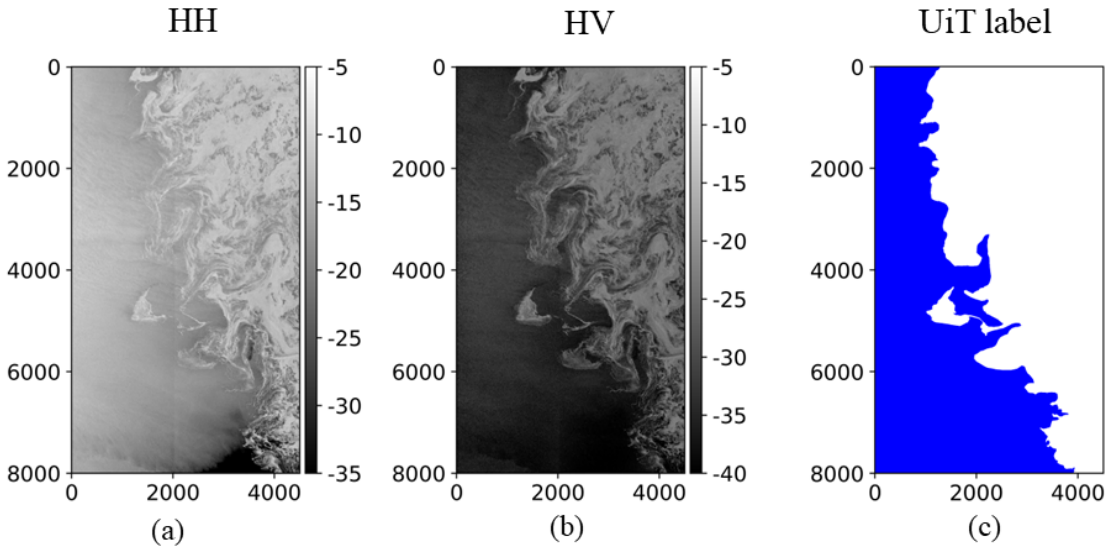

Fig. 2. (a) HH and (b) HV intensity (in dB) for a selected region of an S1 example image (scene NO.11 in Table I), together with (c) corresponding ice–water labels from the UiT dataset. While the ice–water mask captures a fair amount of spatial detail, it does not reflect the pixel-level spatial information that is visible in the SAR imagery, especially in the area close to the ice edge.

datasets with less spatial detail (Figs. 1 and 2). Hence, less human intervention in training set generation is desirable.

Data augmentation is a method to increase the quantity of training samples without human intervention. It is a fairly new methodology, which has proven successful to overcome the challenge of small datasets [18] or datasets with imbalanced class abundances [19], [20]. Such imbalance can be a problem in the case of ice-type mapping, where sea ice and water may not be evenly distributed or the number of pixels for various ice types may vary. Data augmentation also reduces the overfitting phenomenon and increases the generalization ability of CNNs. Hence, it provides an alternative option in cases when a small labeled dataset already exists. Many researchers have explored various data augmentation methods. For instance, operations, such as rotating, flipping, enlarging the contrast among pixels, and blurring have been widely implemented in computer vision-related tasks and were adopted by Khaleghian et al. [13] for sea ice applications. Zhang et al. [21] utilized the convex combination of multiple samples and labels to create new samples, increasing the diversity of the sampling and the robustness of the model performance. Furthermore, variants of generative adversarial networks (GANs) have been proposed by Salimans et al. [22] and have been adopted to data augmentation for imbalanced datasets via transformation [23], [24]. Apart from the above methods for data augmentation, a number of researchers have utilized physics-based data augmentation in training deep learning networks [25], [26], [27].

Motivated by the aforementioned works, we propose in this article a new physics-based data augmentation approach for extending the training data used for training a DNN to separate sea ice and open water at the individual SAR pixel level. In the model, we take advantage of the physics that the IA dependence of the radar backscatter intensity is different for sea ice and open water, which is being demonstrated by several research papers [9], [10], [28].

This article is organized as follows. Section II describes the selected study area, as well as the utilized dataset. Section III presents the methodology, including the training data generation with physics-based augmentation, the CNN model architecture, and the configuration of the training process. Section IV presents the performance of the proposed augmentation approach by depicting classification results, with error metrics of the model, as well as comparison with other ice chart products. Section V discusses the problem of noisy pixels over transition areas in the scenes as well as the open water inference. Meanwhile, possible future solutions to these issues are proposed. The main findings are concluded in Section VI, giving several perspectives on the current model performance and further steps to take in future work.

II. Study Area and Satellite Data

A. Study Area

In this study, we focus on the geographic region between Greenland and Svalbard (22°W–25°E, 69°N–83°N). The air temperature at 2 m above the surface increases from March to June and then decreases afterward. The lowest temperature can reach −39.1 °C on March 10, while the temperature reaches at a peak of 13.4 °C on June 30 according to Danmarkshavn station's measurements carried out by DMI over 2020 (https://confluence.govcloud.dk/display/FDAPI).

This region is characterized by large open water areas as well as variable and highly dynamic sea ice conditions. From East to West (Svalbard to Greenland), the region generally shows a transition from predominantly open water conditions close to Svalbard, via a marginal ice zone (MIZ) with SIC between 0.2 and 0.8, toward dense drift ice and finally landfast sea ice close to the East Greenland coast. The landfast ice area largely consists of rough and strongly deformed rubble fields, which are built up from drift ice that was locked in place and became landfast. These rubble fields are intersected by

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regions of very smooth and level landfast ice, which formed in situ under wind- and wave-protected conditions. In the drift ice area, a mix of first-year and multiyear ice can be found, intersected by ridges and rubble fields as well as leads that can either show open water conditions or can be covered by thin nilas or young ice. The open water area closer to Svalbard provides a large fetch for wind and can therefore show variable SAR backscatter intensity due to changing sea surface state caused by varying wind and ocean current conditions. Overall, with its diverse sea ice conditions in combination with its strategic importance for Arctic operations, the selected area provides a suitable and challenging test region for our present study.

B. Satellite Data

1) Sentinel-1 SAR Data: Sentinel-1 is an SAR mission with global observation coverage and is part of the European Space Agency's (ESA) Copernicus program. It currently consists of two satellites, with Sentinel-1A launched on 3 April 2014 and Sentinel-1B on 25 April 2016. Each individual satellite has a 12-day repeat cycle at the equator, which results in a combined six-day repeat cycle for both satellites. The revisit frequency increases from low to high latitudes, resulting in a temporal resolution of 1–3 days in Arctic regions. It should be noted that the Sentinel-1B mission ended on 23 December 2021 due to a failure in the instrument electronics power supply. Sentinel-1 data are freely available and can, for example, be downloaded via the Copernicus open-access hub (https://scihub.copernicus.eu/).

Sentinel-1 operates at the C-band frequency and offers four exclusive acquisition modes with different swath widths: stripmap (SM), interferometric wide (IW) swath, extra-wide (EW) swath, and wave (WV) mode. In this study, we use EW data in ground range detected (GRD) format at medium resolution. The EW mode is the standard observation mode over open water and has a swath width of approximately 410 km. It covers roughly 10 000 × 10 000 pixels in total with each pixel corresponding to approximately 40 × 40 m on the ground. Meanwhile, the IA across the swath ranges from 18.9° at the near range to 47.0° at the far range. Several studies have shown that the noise floor has a strong influence on Sentinel-1 observations of sea ice [29]. To mitigate noise effects, we apply the denoising procedure proposed by the NERSC [16]. After noise correction, we clip the HH and HV intensity values to intervals ranging from −30 to 0 dB and −35 to −5 dB, respectively, to remove the outliers.

Each Sentinel-1 scene can be uniquely identified by its name string. For the easier readability of this article, we use a running ID number for all S1 images that we explicitly refer to in the text. A list that matches the running IDs with the unique name strings of the corresponding S1 images is given in Table I.

In this work, eight S1 EW images acquired between January and March 2021 (see NO.1–8 of Table I for scene IDs) are selected to generate the training data for later usage (see green rectangles in Fig. 3). The selection of these images is based on the criteria that they should cover different geolocations, varying sea ice conditions, as well as both ascending and descending image acquisitions. The footprints of all training images as well as all images used for inference and validation in this study are shown in Fig. 3.

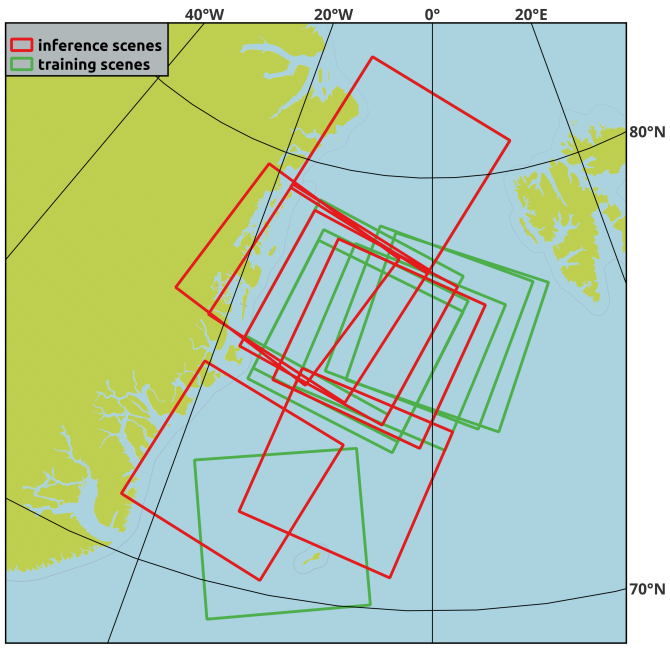

Fig. 3. Overview of all image footprints used for either training (green) or inference (red) in this study.

2) Sentinel-2 Optical Data: Like Sentinel-1, Sentinel-2 is part of ESA's Copernicus program and currently consists of two satellites, Sentinel-2A (launched on 23 June 2015) and Sentinel-2B (launched on 7 March 2017). Both satellites carry a MultiSpectral Instrument (MSI) that samples 13 spectral bands, with the central wavelength of the individual bands ranging from 443 to 2190 nm at a spatial resolution of 10, 20, or 30 m. Sentinel-2 provides a swath width of 290 km and the twin satellites give a combined revisit frequency of five days at the equator. Detailed configuration can be found via https://sentinels.copernicus.eu/web/sentinel/technical-guides/sentinel-2-msi/msi-instrument, and data can freely be downloaded via the Copernicus open-access hub.

Operating in the visual and near-infrared part of the electromagnetic spectrum, Sentinel-2 is affected by natural sunlight and cloud conditions. It is therefore not suitable as a primary sensor for operational sea ice charting but can be highly useful as complementary data. In this study, we use Sentinel-2 data as auxiliary data to evaluate and judge the performance of our deep learning algorithm, which is based purely on Sentinel-1 imagery.

III. Method

In this section, we describe the details of our study method. We first give a short and concise overview of the experiment setup and then describe the details of the different experiments, including training data selection and data augmentation, model structure selection, hyperparameter settings, and performance evaluation metrics.

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TABLE I
S1 SCENE IDS AND CORRESPONDING INTERNAL SCENE NUMBER IN THIS ARTICLE

scene number	purpose	scene ID
1	train	S1B_EW_GRDM_1SDH_20210101T074539_20210101T074639_024958_02F86C_D497
2	train	S1A_EW_GRDM_1SDH_20210102T073823_20210102T073923_035956_043634_DE5A
3	train	S1A_EW_GRDM_1SDH_20210104T072154_20210104T072254_035985_043735_D4FF
4	train	S1B_EW_GRDM_1SDH_20210105T071250_20210105T071350_025016_02FA36_6592
5	train	S1B_EW_GRDM_1SDH_20210209T175419_20210209T175519_025533_030AD7_84BD
6	train	S1A_EW_GRDM_1SDH_20210301T070437_20210301T070537_025818_031425_6693
7	train	S1B_EW_GRDM_1SDH_20210304T072924_20210304T073024_025862_03159B_D26A
8	train	S1B_EW_GRDM_1SDH_20210309T073735_20210309T073835_025935_0317F0_F5F5
9	evaluation	S1A_EW_GRDM_1SDH_20190110T183531_20190110T183635_025419_02D0D0_E2AC
10	evaluation	S1B_EW_GRDM_1SDH_20180421T080246_20180421T080346_010579_0134BA_95A0
11	evaluation	S1A_EW_GRDM_1SDH_20181016T072958_20181016T073058_024158_02A460_DA8F
12	evaluation	S1B_EW_GRDM_1SDH_20190110T080052_20190110T080152_014429_01ADC3_DDBE
13	evaluation	S1A_EW_GRDM_1SDH_20190417T075536_20190417T075636_026827_0303EB_ADF6
14	evaluation	S1A_EW_GRDM_1SDH_20190717T074617_20190717T074717_028154_032E13_D688
15	evaluation	S1A_EW_GRDM_1SDH_20191019T080241_20191019T080341_029525_035C27_393
16	evaluation	S1A_EW_GRDM_1SDH_20220404T081913_20220404T082013_042621_0515B_D75F
17	evaluation	S1B_EW_GRDM_1SDH_20200601T072921_20200601T073021_021837_02972D_1C76
18	evaluation	S2B_MSIL1C_20220404T143749_N0400_R039_T29XMH_20220404T151309

A. Experiment Setup

We investigate three scenarios for different training datasets.

1) *Scenario 1:* Full training data.
2) *Scenario 2:* 10% of the original training dataset.
3) *Scenario 3:* Expand the 10% of the original training dataset by using our physics-based data augmentation.

In scenario 1, we use the full set of training data, which is generated by manual labeling of the physics-aware Gaussian mixture model (GMM) [10], [30] clustering results. In scenario 2, we use only 10% of the full training dataset to simulate a situation where just a small number of labeled training samples is available, e.g., an ice analyst has just delineated a few polygons. Finally, in scenario 3, we augment the training set from scenario 2 to investigate whether we are able to achieve similar accuracy as in scenario 1.

Since both scenarios 2 and 3 use only 10% of the original dataset, we are able to carry out ten independent experiments and provide an uncertainty estimation for the algorithm performance. More information about the different scenarios is given in Table II. Meanwhile, the detailed procedure for generating the training data for these three scenarios is described in the following.

B. Training Data Generation

1) IA Dependence of SAR Backscatter: The radar backscatter from any surface is determined by both radar and surface parameters. The most important surface parameters are the dielectric properties of the material and the surface roughness, the main radar parameters include frequency, polarization, and local IA. While frequency and polarization are fixed for a given sensor and imaging mode (in our case, Sentinel-1 EW mode), the local IA varies across the swath. It is known that backscatter intensity decreases with increasing IA, and several studies have shown that this decrease is approximately linear for backscatter intensity in decibel (dB) (e.g., [28], [31], [32]). The linear slope for σ^0_{HH} versus IA is generally larger than the slope for σ^0_{HV} versus IA.

Furthermore, it has been shown that the variation of backscatter intensity with IA is dependent on the surface type [28], [31]. In particular, sea ice and open water can have significantly different slopes, on the order of 0.2 dB/1° for sea ice and up to 0.7 dB/1° for open water [32]. Hence, a good IA correction must take the surface type into account. Lohse et al. [9] and Cristea et al. [10] have proposed classification and clustering methods for sea ice classification that include the per-class IA dependence.

2) Training Data Generation Stage 1—GMM Clustering: Cristea et al. [10] proposed an automatic and unsupervised segmentation algorithm for SAR image clustering by integrating the IA dependence. Each cluster is described by a "Gaussian tube" in the intensity-IA space, as the mean value of the Gaussian distribution decreases linearly with IA. A GMM is sequentially applied, using the expectation–maximization (EM) algorithm, with an increasing number of clusters. After each EM convergence, the goodness-of-fit (GoF) of the individual clusters is tested using Pearson's chi-squared test and the least well-fitting one is split. When all clusters are considered good-fits or a maximum number of clusters is reached, the algorithm stops. The final hard cluster labels are assigned according to their associated maximum posterior probabilities. Several parameters can be tuned to affect the final segmentation result. In our work, we used 0.99 as the confidence level, the maximum number of clusters is 15, subsampling to 9000 samples during the iterations, as this usually achieved around ten clusters that were considered simple enough to interpret. These unsupervised segmentation clusters are manually labeled as either "sea ice" or "open water" with the help of expert analysis, visual interpretation, and interpretation of the slope values per cluster.

An illustrative example of this clustering and labeling process is given in Figs. 4 and 5. Fig. 4 shows HH and HV backscatter intensity together with the corresponding GMM segmentation result and the manually assigned cluster labels for an S1 scene captured on 2021 January 4. For this scene, 15 total clusters are obtained by the GMM, although only the main four are easily visible. In Fig. 5, the HH intensity

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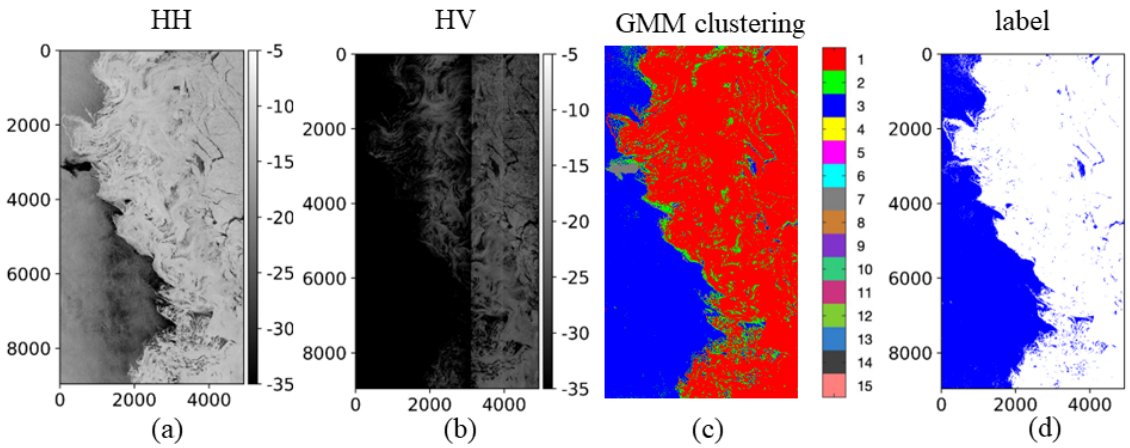

Fig. 4. S1 example image (scene NO.3 in Table I), showing (a) HH and (b) HV backscatter intensity (in dB), and (c) GMM segmentation result and (d) corresponding classes after manually labeling segments as sea ice (white) or open water (blue).

in dB is shown against IA for three main clusters and the corresponding IA linear slopes (obtained by linear least-square fitting) are identified as well. Both the visual interpretation of the intensity images in Fig. 4 and the interpretation of the IA linear slope steepness in Fig. 5 lead us to the labeling of the green and the red classes as sea ice and the blue class as open water.

As we are targeting binary ice–water classification, separate ice clusters [see Fig. 4(c)] are merged into one single sea ice class. Similarly, different open water clusters [see Fig. 4(c)] are merged into one single open water class [Fig. 4(d)]. The criterion for merging classes is mainly based on: 1) the linear slope [shown in Fig. 5] where ice has a shallow linear slope while water has a steep linear slope and 2) further consulting with ice experts. Meanwhile, the IA linear slopes of all the clusters as found by the segmentation algorithm are recorded independently. For the following steps, the segmentation results are further subdivided into patches of size of 224×224 pixels (roughly 9×9 km) in our study. The patch size of 224×224 pixels is selected as this patch size is used by most CNN-based models (for instance, AlexNet developed by Krizhevsky et al. [33], VGG16 proposed by Simonyan and Zisserma [34], and Resnet developed by He et al. [35]).

3) Training Data Generation Stage 2—IA Linear Slope-Based Data Augmentation: In the next step, the individual IA linear slopes captured in stage 1 of the training data generation are utilized to augment the training dataset. For this purpose, we define a transformation function that adjusts the IA values of the original reference patch to any target IA range between 19° and 47°. Taking the original patch labels, clusters, and cluster-dependent IA linear slopes into account, each pixel's backscatter values are then projected along the corresponding slope to the new IA value. Since this transformation makes use of the physics-based principle of per-class backscatter variation with IA, we refer to it as physics-based data augmentation. The augmentation is defined by the following equations:

$$\mathrm{IA}_{i,j}^{\mathrm{new}} = \mathrm{IA}_{i,j}^{\mathrm{ref}} + m * \mathrm{int} \tag{1}$$

$$\mathrm{HH}_{i,j}^{\mathrm{new}} = \mathrm{HH}_{i,j}^{\mathrm{ref}} + \left(\mathrm{IA}_{i,j}^{\mathrm{new}} - \mathrm{IA}_{i,j}^{\mathrm{ref}}\right) * S_{\mathrm{HH}}^{k} \tag{2}$$

$$\mathrm{HV}_{i,j}^{\mathrm{new}} = \mathrm{HV}_{i,j}^{\mathrm{ref}} + \left(\mathrm{IA}_{i,j}^{\mathrm{new}} - \mathrm{IA}_{i,j}^{\mathrm{ref}}\right) * S_{\mathrm{HV}}^{k} \tag{3}$$

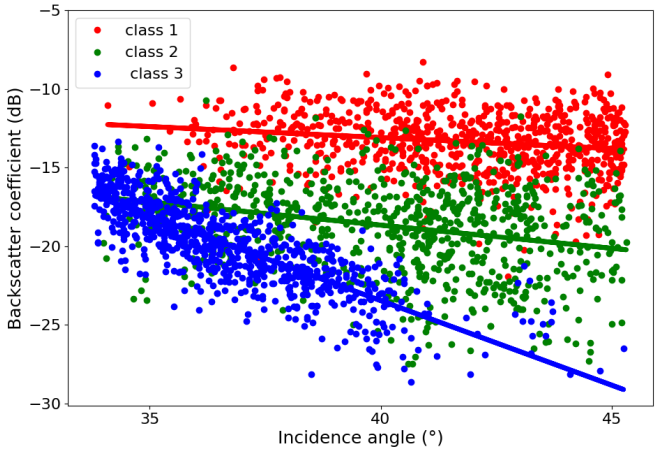

Fig. 5. Example of slopes of backscatter intensity with IA for different clusters. The clusters correspond to the GMM segmentation shown in Fig. 4(c) (here, red and green indicate different ice classes, while blue is the water class).

where int is the predefined IA interval (here, we choose 1°) and m is the multiplication factor for the shifting the IA values. $\mathrm{IA}_{i,j}^{\mathrm{ref}}$ and $\mathrm{IA}_{i,j}^{\mathrm{new}}$ represent the IA of the reference pixel and its corresponding pixel after the augmentation, respectively. Similarly, $\mathrm{HH}_{i,j}^{\mathrm{ref}}$ (and $\mathrm{HV}_{i,j}^{\mathrm{ref}}$) and $\mathrm{HH}_{i,j}^{\mathrm{new}}$ (and $\mathrm{HV}_{i,j}^{\mathrm{new}}$) represent the backscatter intensities of the reference pixel and its corresponding pixel before and after augmentation, respectively. S_{HH}^{k} and S_{HV}^{k} denote the IA linear slopes of the cluster k to which the pixel (i, j) belongs. In Fig. 6, a sample for the original reference patch and new patch after physics-based augmentation is illustrated.

We perform the proposed two-stage training data generation routine for eight S1 EW images acquired between January and March 2021. The 500 boundary pixels in the upper, lower, left, and right edges of the scenes have been removed before the procedure, as these pixels are more likely to be affected by noise or boundary effects. In the end, we are able to generate 5910 patches for stage 1, based on the aforementioned procedure of GMM clustering and expert analysis of the clusters. Following the 80% versus 20% rule for training and validation separation, we randomly select 4731 patches for training and 1179 for validation, as shown in Table II.

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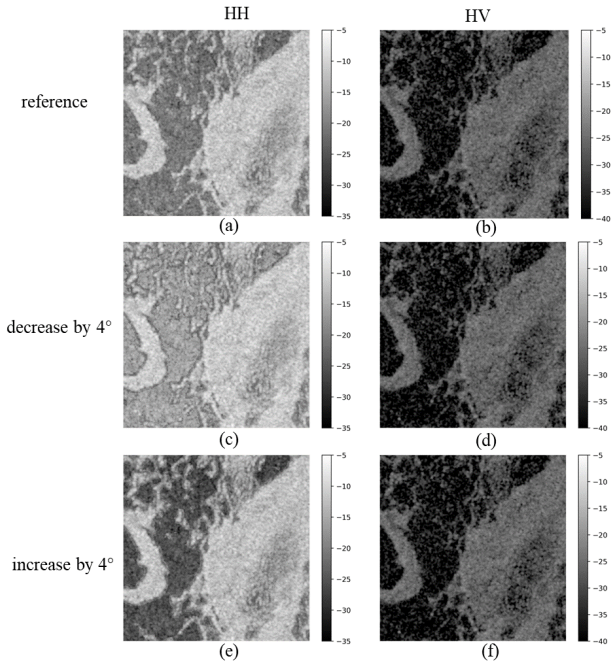

Fig. 6. Example of a 224 × 224 pixel patch before and after physics-based augmentation. The top row shows backscatter intensities ((a) for HH, (b) for HV) from the original image (stage 1 training generation). Rows 2 and 3 show backscatter intensities ((c), (e) for HH and (d), (f) for HV) after data augmentation (stage 2 training generation). Backscatter values are projected along the respective slopes of their corresponding clusters to IA values with 4 degrees below (row 2) and 4 degrees above (row 3) the reference IA.

The patches allocated in the training set are used to learn the hyperparameters of the classifier, whereas the validation data are used to independently assess its performance.

This dataset is being considered the complete dataset and will be used for training in our scenario 1 experiment. Meanwhile, to simulate a situation with fewer available training patches, we split the 4731 training patches into ten equal-sized partitions such that each partition consists of 470 training patches. Taking each of these ten subsets in combination with the complete 1179 patches for validation, we generate ten training and validation dataset couples. These form the basis for our experiments in scenario 2.

Finally, we apply the physics-based data augmentation on the ten training sets from scenario 2 to generate the training sets for scenario 3. Hereby, we are able to generate a large amount of training data and make sure that all example training classes are evenly distributed across the whole IA range of the S1 EW mode. We are able to generate approximately 11 000 training patches for each training subset by moving the patches at 1° steps within the IA range between 19° and 47°. The resulting ten enriched training and validation patch couples are the basis for our scenario 3 training configuration. The detailed information for three scenarios is listed in Table II.

C. Model Structure Selection

Various models for semantic segmentation applications are being developed in the community, including FCN developed by Long et al. [36], U-Net [37], Deeplabv3 [38], and their variants, including U-Net++ [39] and deeplabv3+ [40]. Among these methods, FCN normally shows the worst performance compared with others since it employs the most simple architecture with only two or three levels of upsampling, which might not be able to capture enough details. Similarly, Deeplab V3 and Deeplabv3+ only implement one or two upsampling levels, which is sufficient for images with large homogeneous areas. However, in the case of SAR imagery, pixel backscatter values often vary strongly. Hence, using U-Net or its variants appears to be most beneficial. As our starting point for selecting the model architecture, we, therefore, choose the U-Net, which has more encoder (downsampling) and decoder (upsampling) layers.

U-Net employs a symmetric encoder–decoder structure in which the encoder part is trying to extract the features in various levels and the decoder part is utilized to reconstruct the segmentation masks based on integrating the features in different levels. The encoder part can be divided into four similar blocks where each block consists of a sequence of two 3 × 3 convolutional layers followed by a batch normalization (BN) procedure and the rectified linear unit (ReLU) activation function is added in the end. Following these, max-pooling operations with 2 × 2 window are used for downsampling to reduce the feature size. Correspondingly, in the decoder part, a bilinear upsampling operation is employed. In the last step, the kernel size is set as 1 × 1 to ensure that it has the same size as the original image. The original number of filters is 64, 128, 256, and 512 for the four levels. Using this architecture as a starting point, we further designed a simplified U-Net for our study by reducing the number of layers as well as the filters from the original U-Net, which leads to a three-layer U-Net with the numbers of 32 128 256 filters in the three levels. The detailed architecture for our simplified U-Net architecture is shown in Fig. 7. Note that we utilize three channels (HH, HV, and IA) as input into the modified U-Net.

D. Model Performance Evaluation Metric

Various metrics can be considered to evaluate the performance of separation of sea ice and open water. For instance, Park et al. [12] utilized the confusion matrix for the results of their random forest classifier, while Stokholm et al. [15] adopted R^2 to assess the accuracy of their output SIC maps. In this article, we will follow the commonly used error metrics in the field of semantic segmentation, which is mean intersection over union (MIoU). The computation of MIoU is given in the following equations, where k is the number of classes and p_{ij} is the element of the confusion matrix at row i and column j:

$$\mathrm{IoU}_i = \frac{p_{ii}}{\sum_{j=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}} \tag{4}$$

$$\mathrm{MIoU} = \frac{\sum_{i=1}^{k} \mathrm{IoU}_i}{k}. \tag{5}$$

E. Hyperparameter Settings

For all the scenarios, the same configuration of the hyperparameters is implemented for the training process of the simplified UNET. The detailed configuration for the training

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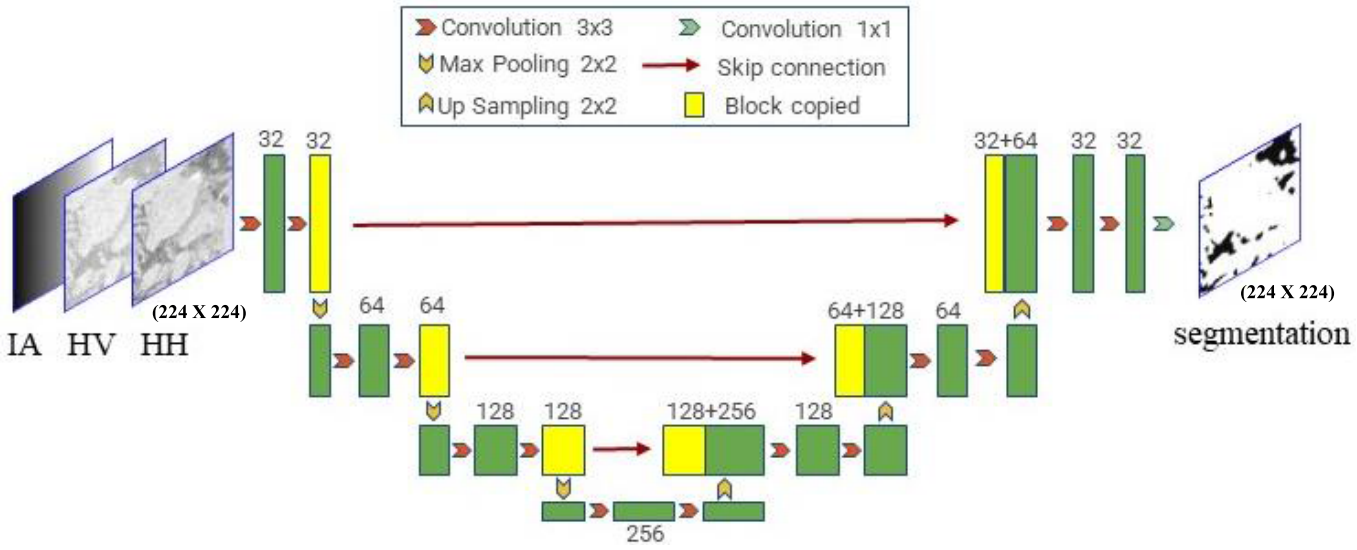

Fig. 7. Illustration of the modified U-Net architecture for sea ice/water segmentation used in this study.

is: the learning rate is set to $1e^{-4}$, the maximum training epoch is set to 200, and the batch size is configured with 64, which is the maximum size for the environment [NVIDIA Quadro RTX 5000 graphic processing unit (GPU)]. Note that we utilize the focal loss [41] to compensate the class imbalance as well as the difference of difficulty to distinguish sea ice and open water classes. In particular, the parameters α and γ are set to 0.9 and 2, respectively. Using this configuration, the training procedure takes 181, 31, and 396 min for scenarios 1–3, respectively (shown in Table II). The complete processing and inference of a new S1 image take approximately 10 min, most of which is used for feature extraction and cropping the of the image into 224 × 224 pixel patches. The actual inference of these patches takes only approximately 20 s.

TABLE II

SETTINGS AND MIOU (CALCULATED BASED ON THE VALIDATION PATCHES) FOR THE DIFFERENT SCENARIOS

scenario		time ($''$)	training patches	MIoU
scenario 1		10847	4731	**94.29%**
scenario 2	1	1879	473	92.30%
	2	1868	473	91.72%
	3	1853	473	92.12%
	4	1855	473	92.27%
	5	1833	473	92.18%
	6	1810	473	92.29%
	7	1809	473	92.28%
	8	1821	473	92.71%
	9	1828	473	92.32%
	10	1830	474	92.28%
	mean			92.25 ± 0.24%
scenario 3	1	23695	10871	92.62%
	2	23898	10988	92.69%
	3	23875	11001	92.82%
	4	23939	11020	92.79%
	5	23562	10781	92.96%
	6	24155	11074	92.48%
	7	23935	10970	92.79%
	8	23929	11015	92.84%
	9	23706	10821	92.49%
	10	22668	10352	92.83%
	mean			92.73 ± 0.16%

IV. RESULTS

A. General Model Performance

The performance results (based on the validation patches) of the three different scenarios are summarized in Table II. Recall that we have split the complete training set and carried out ten experiments each for scenarios 2 and 3. Clearly, the highest accuracy is achieved in scenario 1, where we use the complete training set. Furthermore, we find that the physics-based data augmentation of training data improves the classification accuracy. This is indicated by the MIoU improved from 91.72%–92.71% to 92.48%–92.96%. Meanwhile, the standard deviation for the scenario 2 results is 0.24% for MIoU, while the standard deviation of MIoU for scenario 3 is 0.16%. The lower variance for scenario 3 accuracy compared to scenario 2 accuracy indicates that the model performance is more robust and stable with scenario 3, due to the benefit of physics-based data augmentation. The overall higher score achieved in scenario 1 can be attributed to the fact that the reduced training data of scenarios 2 and 3 cover fewer and less diverse sea ice situations than the full training of scenario 1. Hence, while physics-based data augmentation improves the results, it cannot fully substitute the manual labeling of additional training data.

To illustrate changes during the training process of the model, we plot the evolution of error metrics of the training for scenario 1 together with experiment 5 of scenarios 2 and 3 in Fig. 8. The top [Fig. 8(a)] shows the change of MIoU with training epochs, and the bottom [Fig. 8(b)] shows the loss evolution with training epochs. It can be seen that when using physics-based augmentation, the loss for the training and validation process drops faster compared to the training without data augmentation. Furthermore, the training converges to 0.018 when we include physics-based data augmentation (experiment 5 of scenario 3), which is close to the value of

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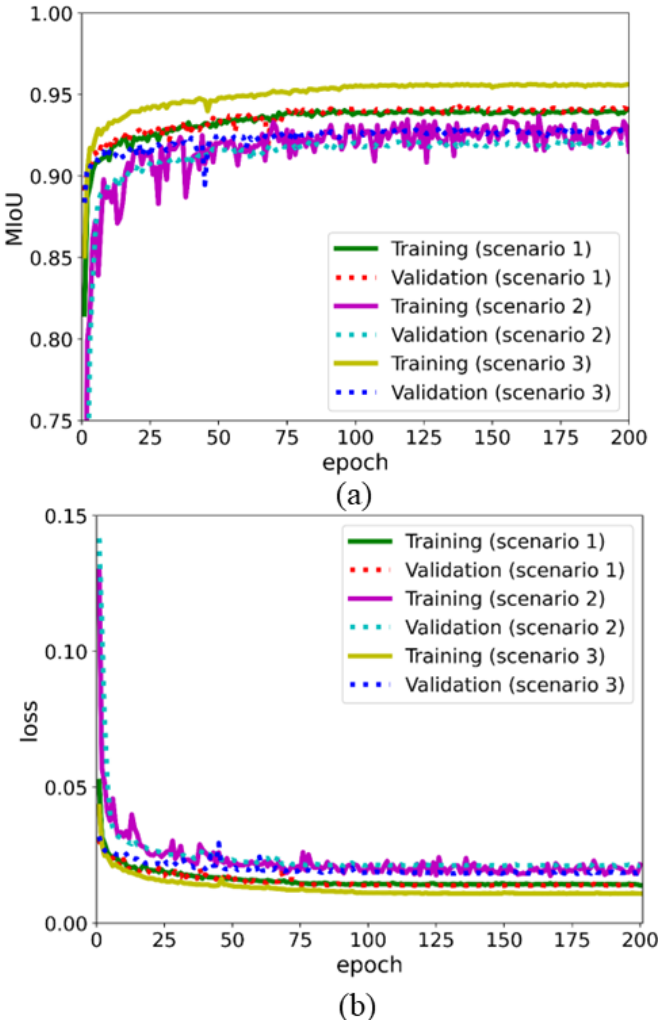

Fig. 8. Evolution of (a) MIoU and (b) loss for training and validation of different scenarios (scenario 1, experiment 5 of scenario 2, and scenario 3).

0.014 when using the complete dataset (scenario 1) and better than the convergence of experiment 5 from scenario 2, which converges to 0.021. This shows that we are able to achieve a better performance during training by including physics-based data augmentation. Finally, the training process of scenario 3 is stabilized at the epoch of 130–140, which is the same as using the complete dataset in scenario 1, but much earlier than for the training without data augmentation in scenario 2, which stabilizes at the epoch of 170–180.

B. Comparison With CAS Product

To further evaluate the performance of our model both with and without physics-based data augmentation, we compare our model inference with other existing products. Several studies, for example, Stokholm et al. [15] or Andersson et al. [42], generate automated ice charts that provide SIC at a rather coarse resolution (in the order of kilometers). These products are difficult to use for comparison with our model inference since the spatial resolution of our inference is 40 m. We, therefore, use the sea ice mapping produced by Wang and Li [17] from the Chinese Academy of Sciences (CAS) for the comparison in this study. The CAS dataset is based on an ensemble prediction of multiple U-Net models. Its spatial resolution is 400 × 400 m, resulting from a 10 × 10 pixel averaging window that was applied during processing. In contrast, our algorithm provides output labels at the original S1 EW pixel level of 40 × 40 m. Meanwhile, the CAS dataset is based on training from 167 S1 scenes distributed over all seasons, while we use only 8 S1 scenes acquired during winter conditions.

A visual comparison of the inference results for four different S1 images (scene NO.12–15, Table I) is shown in Fig. 9. Note that we compare with the results from our model in scenario 1 (full dataset), as well as one selected subset from scenarios 2 and 3 (experiment 5). Overall, our model inference matches well with the CAS classification. However, several differences in the inference can be found, some of which are highlighted by colored rectangles in Fig. 9.

For example, in all four examples shown here, artifacts from the patch edges (224 × 224 pixels) are visible in the inference from scenario 2 [red rectangles in Fig. 9(d), (j), (p), and (v)]. At these edges, open water is misclassified as sea ice. The issue is resolved in the inference from scenario 3, where physical data augmentation is applied.

Furthermore, there is considerable misclassification of open water as sea ice at the boundary between the EW1 and EW2 swaths of the S1 images, highlighted by the purple rectangles in Fig. 9(d), (e), (j), and (k). This is a commonly known issue in ice–water classification caused by the noise properties of S1 imagery [9] and has already been recognized as problematic by different researchers, who suggested several methods to address this issue [16], [43], [44]. As can be seen in Fig. 9(c) and (i), the CAS algorithm is able to classify correctly around the swath boundary, while both our models from scenario 2 [Fig. 9(d) and (j)] and scenario 3 [Fig. 9(e) and (k)] output misclassified pixels. Hereby, we explore the potential of adding more training samples to overcome these noise issues. Instead of adding new training patches by physics-based data augmentation, we can also retrain our model using the complete dataset (scenario 1), shown in the last column of Fig. 9. It can clearly be seen that using the complete training set that contains more diverse sea ice situations, the noisy pixels in the transitional area can be removed significantly [see purple rectangle in Fig. 9(f) and (l)]. It appears that in this case, the physics-based data augmentation cannot improve the inference as much as additional, completely new, and independent training labels. Meanwhile, the remaining noisy pixels across the swath boundary can be easily removed by applying a spatial filter. Generally, better noise estimation as well as denoising methods by using physical models or deep learning models could be considered in the future. It can also be seen in Fig. 9(f) that using the complete training set (scenario 1) enables us to delineate the ice edge even better (see purple rectangle). The resulting ice edge matches better with the edge from the CAS dataset [Fig. 9(c)].

Thus, we conclude that a diverse training set is essential to resolve noise-related classification errors for the separation of sea ice and open water. Furthermore, postprocessing techniques to clean the false alarms after inference are recommended.

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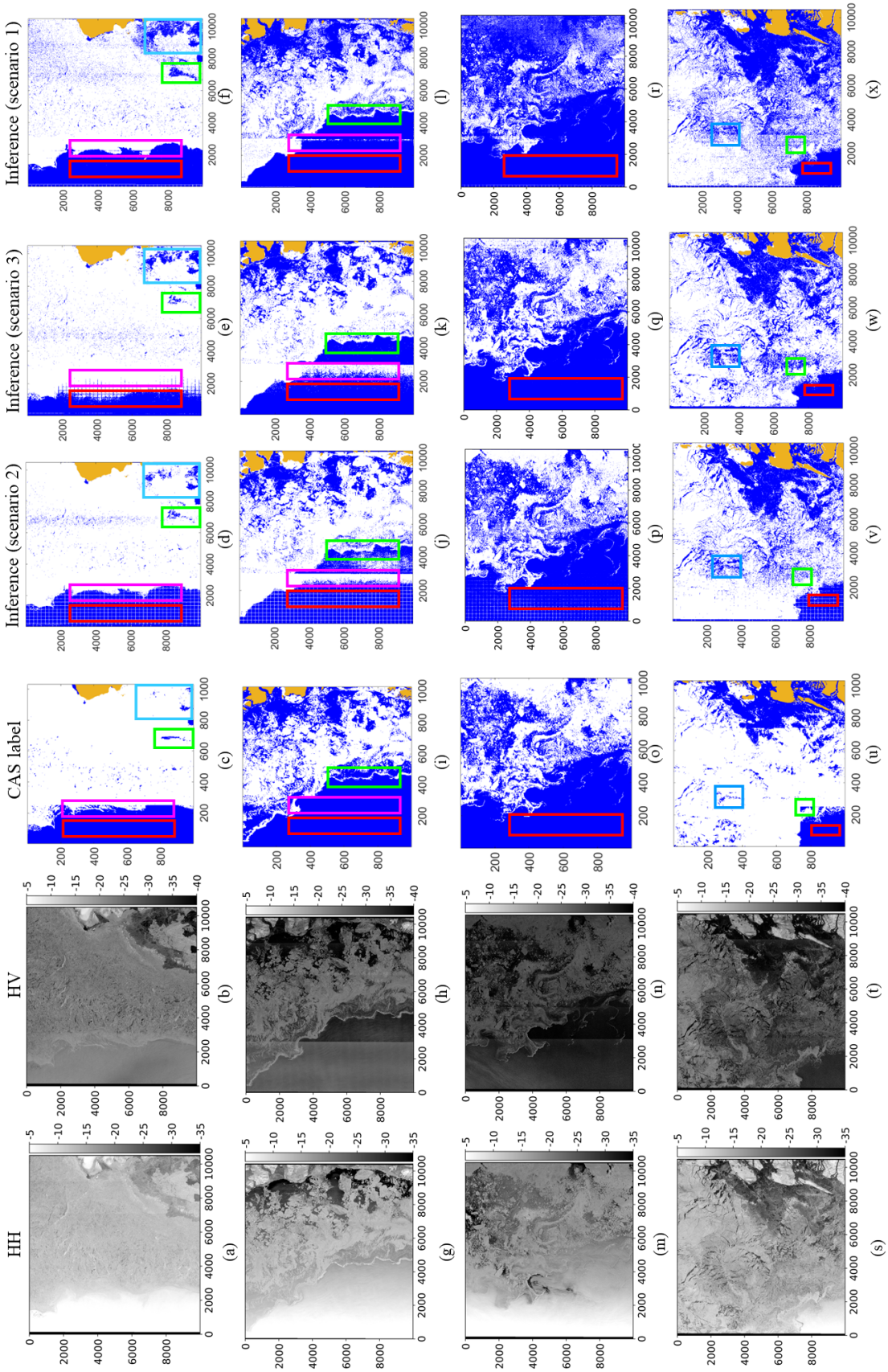

Fig. 9. From left to right: HH((a),(g),(m),(s)), HV((b),(h),(n),(t)), CAS labels((c),(i),(o),(u)), model inference (scenario 2, (d),(j),(p),(v)), model inference (scenario 3,(e),(k),(q),(w)), and model inference (scenario 1,(f),(l),(r),(x)) for four S1 example images (scene No. 12–15, Table I). Note the coarser spatial resolution of the CAS data set (400 m compared to 40 m). Regions of particular interest are highlighted by colored rectangles (artifacts from the patch edges are identified by the red rectangle, purple box indicates the transition are between swath EW1 and EW2, green and blue boxes indicate other areas which are being identified differently by CAS label and our inference) and referred to in the text. Water is blue, ice is white, and land is masked yellow.

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Further differences between the CAS dataset and our model inference are highlighted by the green and blue rectangles in the 1st, 2nd, and 4th rows of Fig. 9. For example, the green rectangle in the 2nd row of Fig. 9 highlights an area close to the ice edge, which is identified as open water by the Fig. 9(i), while scenario 2 [Fig. 9(j)] and scenario 3 [Fig. 9(k)] inference find more sea ice. Without additional information, such as optical data or in situ ground-truth observations, it is not straightforward to conclude which algorithm is more accurate. The area is characterized by lower backscatter values than the surrounding sea ice, which can be an indication of either relatively calm open water areas or comparatively smooth newly formed sea ice. The discrepancy between the CAS labels and our inference, in this case, may be emphasized by the difference in spatial resolution. Similar considerations apply for the area marked by the green and the blue rectangles in the 4th row of Fig. 9, where our model inference shows slightly more open water compared to the CAS labels. In green rectangle in particular, the CAS labels [Fig. 9(u)] show a small portion of water with a sharp vertical line, whereas our inference from scenarios 2 and 3 shows a smooth ice–water boundary.

Finally, areas of landfast ice close to the East Greenland coast [Fig. 9(c)–(f), blue and green rectangles] prove to be challenging for all presented algorithms. At the time of the image acquisition (January 2019), the highlighted area is entirely covered by landfast sea ice. However, when the landfast ice forms thermodynamically in situ, it will produce a very smooth surface that results in low backscatter intensity of the SAR signal. This can be misinterpreted as calm open water. This issue is shown in more detail in Fig. 10 and Section IV-C.

Overall, our model inference matches well with the CAS labels. Noise issues and transitions at swath boundaries, as well as large areas of young ice and very smooth landfast ice, can cause misclassification. Without additional information or complementary data, it is not always possible to assess which model is more accurate. The CAS model performs better in noisy open water areas, but it requires more computational resources (recall that it is based on ensemble prediction of multiple UNET models) and offers a coarser spatial resolution. Some network-related artifacts (patch boundaries) from our model can be mitigated by physics-based data augmentation.

C. Comparison With Optical Imagery

We also compare the model inference with optical imagery from Sentinel-2. During cloud-free conditions, optical data allow for straightforward separation of sea ice and open water based on image brightness. However, in the MIZ, sea ice drift and differences in image acquisition time make it difficult to use optical images for pixelwise validation of our model inference. In Fig. 10, we, therefore, show an area along the edge of the landfast ice close to East Greenland. The Sentinel-1 SAR image was acquired on 4 April 2022 at 08:19:13 UTC, the optical Sentinel-2 image was acquired at 14:37:49 UTC on the same day.

For most of the images, the model inference is correct. Some errors occur, indicated by the red rectangles in Fig. 10, where very smooth and level landfast ice is misclassified as open water. This classification error was already observed in Section IV-B. While much of the landfast ice in the region is formed in other areas and gets locked in place as highly deformed rubble fields, some of the ice forms in situ under very protected conditions. This results in unusually smooth and undeformed sea ice, which is classified incorrectly by our model. The green rectangles in Fig. 10 mark polynya regions, in which new sea ice is formed. As the overall ice drift is toward the south, the northern part of the polynyas is ice-free, and ice thickness gradually increases in the southern direction within the polynya. The model inference captures the transition between open water and young ice in the polynya very well.

Additional comparisons with more optical data would be desirable, in particular in the MIZ. However, sea ice conditions close to the ice edge are highly dynamic and the structural patterns of the ice change so quickly that ice drift compensation in this area is not feasible. In this section, we, therefore, restrict ourselves to the one example discussed above. For future work, a better approach for more and easier comparison of classification results and optical data close to the ice edge would be beneficial.

V. Discussion

A. Seasonal Variation for Sea Ice and Water Segmentation

Apart from some remaining ambiguities (young ice and very smooth and level landfast ice), the results of this study show that our model is able to correctly classify open water and sea ice during the winter season for both calm and windy water conditions [Fig. 9(g)–(r)]. However, as we only include training data from the winter (January–March), the inference for the melt and summer season is expected to be more challenging. An example image acquired on 1 June 2020 (scene NO.17 in Table I) is shown in Fig. 11, with wrongly classified areas highlighted by colored rectangles.

For this example image, the surface properties of the ice and snow have been altered by warm temperatures, resulting in misclassification of sea ice areas as open water (green and purple rectangles in Fig. 11). In addition, there is considerable misclassification of wind-affected water as sea ice (red rectangle in Fig. 11). Overall, the ice edge is not well delineated. The most straightforward way to obtain a reasonable inference for images acquired during melting conditions is to include similar imagery captured in the summer in the training dataset and retrain the model carefully.

B. Young Ice and Smooth Landfast Ice Areas

As clearly shown in Fig. 10, areas of smooth and level landfast ice are challenging for automated separation of sea ice and open water. Although it achieves slightly better results, this is also true for the CAS algorithm (Fig. 9). While further investigation of this issue is beyond the scope of the present study, there are several approaches that could help to improve the classification of these regions. The high-resolution SAR observations can be merged with low-resolution PM

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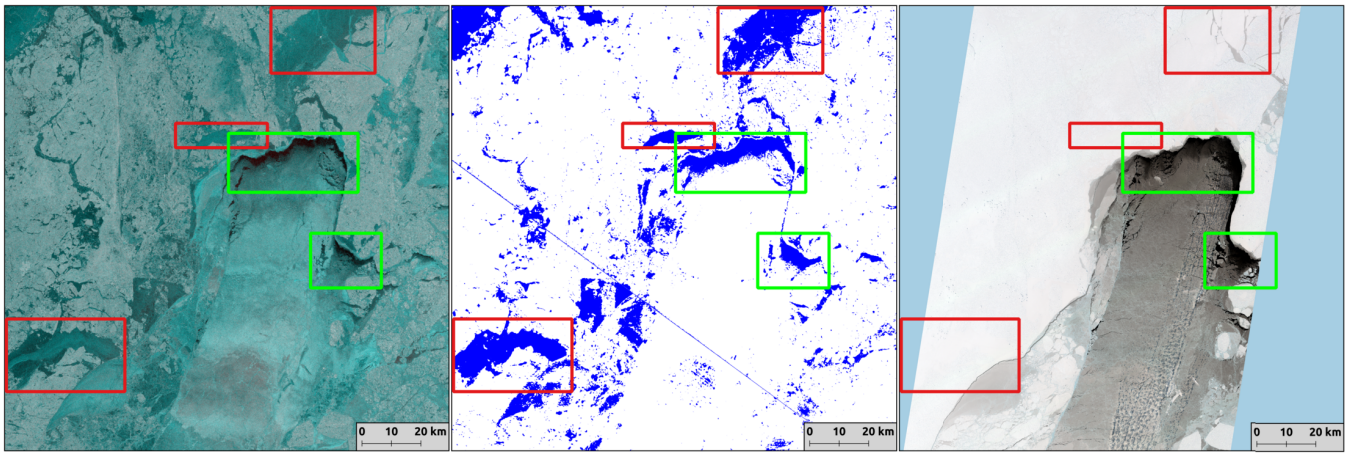

Fig. 10. Comparison of Sentinel-1 (left, false-color composite of R:HV, G:HH, and B:HH), inference (middle) (scene NO.16 in Table I), and Sentinel-2 visual channels (right) (scene NO.18 in Table I). Red rectangles indicate areas where very smooth and level landfast ice close to East Greenland is misclassified as open water. Green triangles indicate areas where the algorithm successfully identifies polynya areas that are opening up and partly covered with young ice.

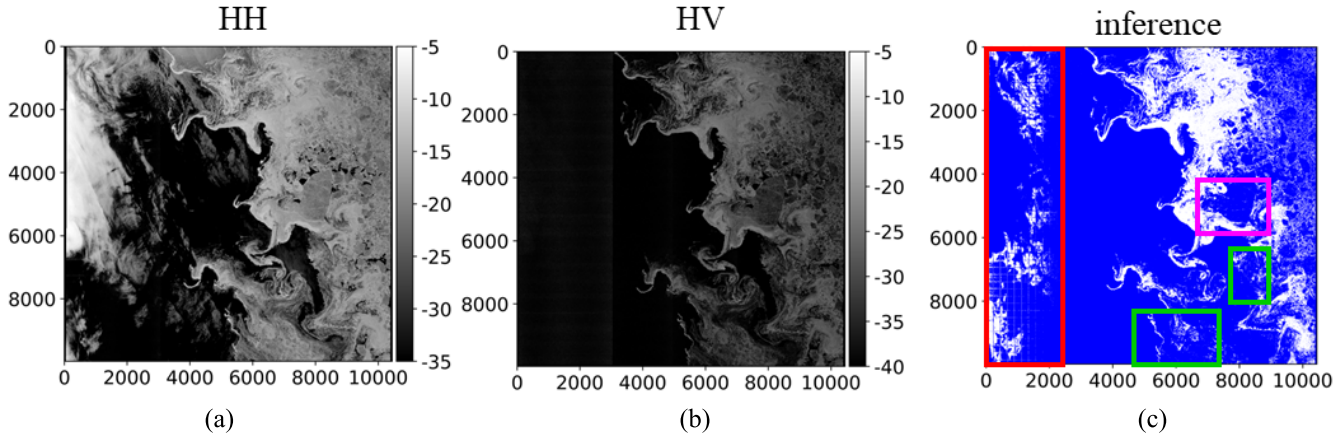

Fig. 11. (a) HH and (b) HV (dB) intensity together with (c) inference result for a selected example image that was acquired in the summer (scene NO.17 in Table I). The colored rectangles in (c) indicate areas with incorrect inference and are referred in the text. Such errors are expected, as the current version of the model is trained on winter scenes only.

observations to a multimodal dataset. PMR is less sensitive to surface roughness and would help to correctly identify landfast ice regions as sea ice. However, merging these different data sources comes with the challenge of their different spatial resolution, as well as differences in acquisition time, which will be problematic in drift ice areas. Alternatively, the SAR-based classification could potentially be combined with ice drift algorithms, which are capable to identify landfast ice areas. Further research in this area is required in future studies.

C. Effect of the Training Size on Physics-Based Augmentation

The main focus of the present study was the investigation of our proposed physics-based data augmentation for classification of sea ice and open water. The results presented in Table II show that the data augmentation leads to a significant improvement in classification accuracy when a small subset of 10% of the full training set is used. However, it remains unclear whether the augmentation is also beneficial when the original dataset is already larger. To investigate this question, we performed our proposed data augmentation repeatedly on differently sized training sets, corresponding to subsets of 30%, 50%, 70%, and 100% of the full training dataset (4731 patches). The resulting performances are shown in Fig. 12 and summarized in Table III.

TABLE III

SETTINGS AND MIOU (CALCULATED BASED ON THE VALIDATION PATCHES) FOR TRAINING WITH DIFFERENT FRACTIONS OF THE FULL DATASET, WITH AND WITHOUT DATA AUGMENTATION

scenario	fraction (%)	time (″)	training patches	MIoU
no augmentation	10	1839	473	92.25%
	30	3855	1458	93.15%
	50	5749	2309	93.86%
	70	7740	3303	94.02%
	100	10847	4731	94.29%
with augmentation	10	20739	10889	92.73%
	30	77231	36315	93.66%
	50	130212	60525	94.00%
	70	177877	84736	94.09%
	100	253255	121051	94.47%

Fig. 12 shows that the performance with data augmentation is always better than without data augmentation. However, the MIoU gain achieved by the data augmentation is larger when only a small fraction of the full dataset is used. For

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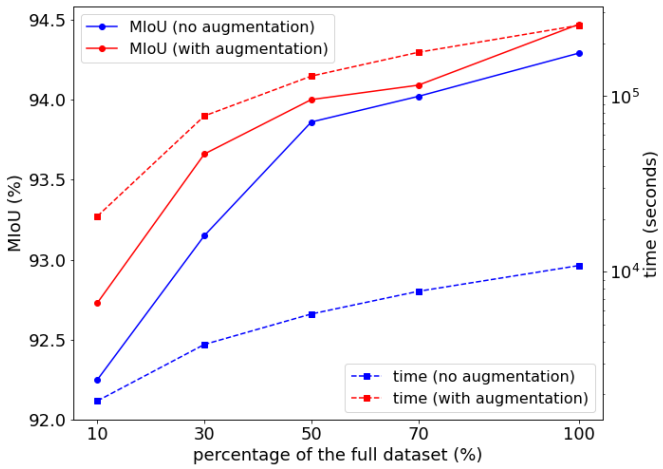

Fig. 12. MIoU and processing time for training with different fractions of the full dataset. (Note that the MIoU is plotted against the first y-axis and processing time (logarithmic scale) is plotted against the second y-axis.)

a larger fraction of the full dataset, the performance gain is smaller. More specifically, the gain in MIoU of training with augmentation using 10% of the full dataset is 0.48% (from 92.25% to 92.73%), while the gain of augmentation with larger training sets, i.e., 50%, 70%, and 100% of the available training data, will result in a gain of 0.14%, 0.07%, and 0.18%, respectively. We also note that when training with the full dataset (100%, i.e., all 4731 patches), the model achieves an MIoU of 94.29%. The training procedure for this case takes approximately 3 h. However, to achieve a similar performance with data augmentation, we need to use already 70% of the full training set. Hence, we need a very large number of training patches, consequently resulting in a long training process, taking more than 49 h. Note the training time (in seconds) on the logarithmic y-axis in Fig. 12. Considering the large amount of computer resources required for such long training processes, we conclude that it is preferable to manually label additional training data, whenever this is possible. However, as the manual labeling is time-consuming for an expert, this is not always a viable option. In such cases, our proposed physics-based data augmentation provides an alternative way to improve the algorithm's performance.

VI. Conclusion

In this study, we have introduced a novel, physics-based data augmentation routine to increase the amount of training data for ice–water separation in SAR imagery. Our proposed method utilizes the per-class dependency of backscatter intensity (in dB) with IA to shift training patches with known class labels to different IA ranges. To test this method, we have implemented a modified U-Net model to classify sea ice and open water in S1 images. We have trained different versions of the model, first using our full training dataset and then using subsets of the full dataset, both with and without data augmentation. Overall, we find that our proposed method for physics-based data augmentation increases the accuracy of the classification. Using a subset training size of 10% of the full training set, we find that MIoU increases by 0.48% (92.73% compared with 92.25%). We also find that including the new data augmentation results in earlier convergence and stabilization during the training phase of the model (at the epoch of 130–140 compared to the epoch of 170–180). Further experiments with larger subsets of the full training set show that the effect of the data augmentation is largest when the amount of original training data is small and decreases with an increasing amount of labeled samples. Finally, it should be noted that the data augmentation leads to a highly increased training time of the model. Meanwhile, the best accuracy overall is still achieved when using the full training set, as it covers the most diverse sea ice situations. Hence, adding additional manual training patches is preferable to data augmentation whenever possible. When the training set is limited or more manual labeling is not possible though our proposed method can help to increase the classification performance.

We tested the overall performance of our model by comparing the inference for several S1 images with corresponding ice–water labels from the CAS sea ice dataset. The comparison shows that our model produces similar results to the CAS dataset, indicating that it can provide detailed separation of sea ice and open water at fine spatial resolution (original 40 × 40 m pixel spacing of the S1 EW GRDM format). While the overall delineation of the sea ice edge corresponds well with the CAS labels, our training process is much less computationally expensive. We also find several limitations of our current model, the most important ones being the misclassification of open water as sea ice at the boundary between the S1 subswaths EW1 and EW2, and the misclassification of very smooth and level landfast ice with low backscatter values as open water. Thus, more effort is required to handle these situations. Suggestions for future improvement of the model include adding more training data for ice situations that are known to be particularly difficult to classify and applying postprocessing filters to smooth results in parts of the image that are affected by noise. Alternatively, different input features (e.g., texture features or intensity ratios) can be integrated into the model for better separating ambiguous classes. In addition, transformer-based network architectures, which can exploit long-range information effectively with the self-attention mechanism, should be further considered. These could be combined with pure CNN-based networks, which are more robust in understanding short-range information due to the intrinsic locality of the convolution operations. For instance, TransUNet [45] and Swin-UNet [46] will be implemented and tested in the future.

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References

[1] M. C. Serreze and W. N. Meier, "The Arctic's sea ice cover: Trends, variability, predictability, and comparisons to the Antarctic," *Ann. New York Acad. Sci.*, vol. 1436, no. 1, pp. 36–53, Jan. 2019.

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[2] B.-M. Kim et al., "Weakening of the stratospheric polar vortex by Arctic sea-ice loss," *Nature Commun.*, vol. 5, no. 1, pp. 1–8, Sep. 2014.

[3] M. Kretschmer, D. Coumou, J. F. Donges, and J. Runge, "Using causal effect networks to analyze different Arctic drivers of midlatitude winter circulation," *J. Climate*, vol. 29, no. 11, pp. 4069–4081, Jun. 2016.

[4] A. M. Pagano and T. M. Williams, "Physiological consequences of Arctic sea ice loss on large marine carnivores: Unique responses by polar bears and narwhals," *J. Exp. Biol.*, vol. 224, no. 1, Feb. 2021, Art. no. jeb228049.

[5] F. Lasserre and S. Pelletier, "Polar super seaways? Maritime transport in the Arctic: An analysis of shipowners' intentions," *J. Transp. Geogr.*, vol. 19, no. 6, pp. 1465–1473, Nov. 2011.

[6] G. Spreen, L. Kaleschke, and G. Heygster, "Sea ice remote sensing using AMSR-E 89-GHz channels," *J. Geophys. Res.*, vol. 113, pp. 1–14, 2008.

[7] J. Comiso and C. Sullivan, "Satellite microwave and in situ observations of the Weddell Sea ice cover and its marginal ice zone," *J. Geophys. Res., Oceans*, vol. 91, no. C8, pp. 9663–9681, 1986.

[8] R. Saldo, M. Kreiner, J. Buus-Hinkler, L. Pedersen, D. M. Hansen, and A. Nielsen, "AI4Arctic/ASIP sea ice dataset—Version 2," Tech. Univ. Denmark, Lyngby, Denmark, 2020, doi: 10.11583/DTU.13011134.v3.

[9] J. Lohse, A. P. Doulgeris, and W. Dierking, "Incident angle dependence of Sentinel-1 texture features for sea ice classification," *Remote Sens.*, vol. 13, no. 4, p. 552, Feb. 2021.

[10] A. Cristea, J. van Houtte, and A. P. Doulgeris, "Integrating incidence angle dependencies into the clustering-based segmentation of SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2925–2939, 2020.

[11] N. Zakhvatkina, A. Korosov, S. Muckenhuber, S. Sandven, and M. Babiker, "Operational algorithm for ice–water classification on dual-polarized RADARSAT-2 images," *Cryosphere*, vol. 11, no. 1, pp. 33–46, Jan. 2017.

[12] J.-W. Park, A. A. Korosov, M. Babiker, J.-S. Won, M. W. Hansen, and H.-C. Kim, "Classification of sea ice types in Sentinel-1 synthetic aperture radar images," *Cryosphere*, vol. 14, no. 8, pp. 2629–2645, Aug. 2020.

[13] S. Khaleghian, H. Ullah, T. Kræmer, N. Hughes, T. Eltoft, and A. Marinoni, "Sea ice classification of SAR imagery based on convolution neural networks," *Remote Sens.*, vol. 13, no. 9, p. 1734, Apr. 2021.

[14] D. Malmgren-Hansen et al., "A convolutional neural network architecture for Sentinel-1 and AMSR2 data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1890–1902, Mar. 2021.

[15] A. Stokholm, T. Wulf, A. Kucik, R. Saldo, J. Buus-Hinkler, and S. M. Hvidegaard, "AI4SeaIce: Toward solving ambiguous SAR textures in convolutional neural networks for automatic sea ice concentration charting," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4304013.

[16] J. Park, J. Won, A. A. Korosov, M. Babiker, and N. Miranda, "Textural noise correction for Sentinel-1 TOPSAR cross-polarization channel images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4040–4049, Jun. 2019.

[17] Y.-R. Wang and X.-M. Li, "Arctic sea ice cover data from spaceborne synthetic aperture radar by deep learning," *Earth Syst. Sci. Data*, vol. 13, no. 6, pp. 2723–2742, Jun. 2021.

[18] M. Olson, A. Wyner, and R. Berk, "Modern neural networks generalize on small data sets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 3623–3632.

[19] R. Blagus and L. Lusa, "SMOTE for high-dimensional class-imbalanced data," *BMC Bioinf.*, vol. 14, no. 1, pp. 106–121, Dec. 2013.

[20] R. Hasibi, M. Shokri, and M. Dehghan, "Augmentation scheme for dealing with imbalanced network traffic classification using deep learning," 2019, *arXiv:1901.00204*.

[21] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[22] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 2234–2242.

[23] A. Antoniou, A. Storkey, and H. Edwards, "Data augmentation generative adversarial networks," 2017, *arXiv:1711.04340*.

[24] G. Mariani, F. Scheidegger, R. Istrate, C. Bekas, and C. Malossi, "BAGAN: Data augmentation with balancing GAN," 2018, *arXiv:1803.09655*.

[25] A. O. Omigbodun, F. Noo, M. McNitt-Gray, W. Hsu, and S. S. Hsieh, "The effects of physics-based data augmentation on the generalizability of deep neural networks: Demonstration on nodule false-positive reduction," *Med. Phys.*, vol. 46, no. 10, pp. 4563–4574, Oct. 2019.

[26] N. Dahiya et al., "Multitask 3D CBCT-to-CT translation and organs-at-risk segmentation using physics-based data augmentation," *Med. Phys.*, vol. 48, no. 9, pp. 5130–5141, Sep. 2021.

[27] Z. Fabian, R. Heckel, and M. Soltanolkotabi, "Data augmentation for deep learning based accelerated MRI reconstruction with limited data," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 3057–3067.

[28] M. Mäkynen and J. Karvonen, "Incidence angle dependence of first-year sea ice backscattering coefficient in Sentinel-1 SAR imagery over the Kara sea," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6170–6181, Nov. 2017.

[29] Y. Sun and X.-M. Li, "Denoising Sentinel-1 extra-wide mode cross-polarization images over sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2116–2131, Mar. 2021.

[30] Q. Wang, A. P. Doulgeris, and T. Eltoft, "Physics-aware training data to improve machine learning for sea ice classification from Sentinel-1 SAR scenes," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 4992–4995.

[31] M. P. Makynen, A. T. Manninen, M. H. Simila, J. A. Karvonen, and M. T. Hallikainen, "Incidence angle dependence of the statistical properties of C-band HH-polarization backscattering signatures of the Baltic Sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 12, pp. 2593–2605, Dec. 2002.

[32] J. Lohse, A. P. Doulgeris, and W. Dierking, "Mapping sea-ice types from Sentinel-1 considering the surface-type dependent effect of incidence angle," *Ann. Glaciol.*, vol. 61, no. 83, pp. 260–270, Dec. 2020.

[33] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[34] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[36] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[37] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[38] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[39] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Cham, Switzerland: Springer, 2018, pp. 3–11.

[40] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoderdecoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[41] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[42] T. R. Andersson et al., "Seasonal Arctic sea ice forecasting with probabilistic deep learning," *Nature Commun.*, vol. 12, no. 1, p. 5124, Aug. 2021.

[43] P. Q. Lee, L. Xu, and D. A. Clausi, "Sentinel-1 additive noise removal from cross-polarization extra-wide TOPSAR with dynamic least-squares," *Remote Sens. Environ.*, vol. 248, Oct. 2020, Art. no. 111982.

[44] A. Korosov, D. Demchev, N. Miranda, N. Franceschi, and J. Park, "Thermal denoising of cross-polarized Sentinel-1 data in interferometric and extra wide swath modes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5218411.

[45] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[46] H. Cao et al., "Swin-Unet: Unet-like pure transformer for medical image segmentation," 2021, *arXiv:2105.05537*.

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